

Remote Sensing

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Doppler Lidar Wind Retrieval Fundamentals:

Radial (line-of-sight) (LOS) velocity measurement principle:

Velocity of the target in respect of the observer (or the radar) is the rate of change of the vector displacement (u, v, w) between the two points. The term "line-of-sight" is defined as the vector projection of the relative velocity between the two points between the target and the observer. ([source, Everything explained today](#))

For the measurement of wind velocity, the best practice is to obtain a vector wind at a given point in space and view that same point from three or more LOS directions. (requires three or more lidar systems to do this practice).

With the assumption of horizontal homogeneity ($\Delta u, \Delta v = 0$) of the wind field over the sensed volume, the lidar beam scanning techniques can be used to determine the vector wind velocity.

Velocity-Azimuth-Display (VAD)

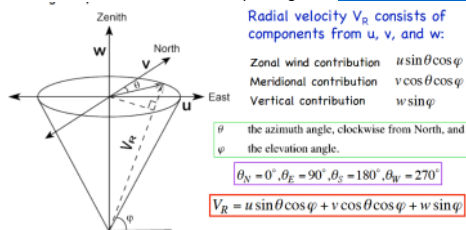
Doppler-Beam Swinging (DBS) ([source, colorado.edu](#))

Scan Strategies:

VAD (velocity-azimuth display):

When the conical scan lidar is at a fixed elevation angle.

The follow calculations for the wind velocity through VAD ([source, colorado.edu](#))



DBS (Doppler beam swing):

Practice of pointing the lidar beam to vertical and then tilted to the east and the tilted to the north.

Through the DBS method three component about the wind vectors can be obtained.

([source, colorado.edu](#))

$$\begin{aligned} u &= -(V_{r2} - V_{r1} \sin \phi) / \cos \phi \\ v &= -(V_{r3} - V_{r1} \sin \phi) / \cos \phi \\ w &= -V_{r1} \end{aligned}$$

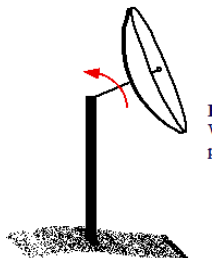
V_{r1}, V_{r2}, V_{r3} are the vertical, east, and north radial velocities

For wind science in application to wind turbines, the VAD method has been reliable to determine the performance of wind turbines (Letizia, 2024).

Staring/RHI:

RHI: Range Height Indicator

The radar has a constant azimuth angle but varies in its elevation angle. The output can be mapped on a vertical plane for atmospheric observations. ([source, uiuc.edu](#))



Primarily used as a method to observe specific weather or storm vertical profiles at a rapid refresh mode. ([source, stromquant.com](#))

Dual-Doppler retrieval geometry for full 3d wind vectors: (Got a bit confused finding a plain definition and methods of this term so I used ecosia ai)

A method of conjoining the atmospheric measurements from two doppler radars that are in different positions looking at the same target. This is done because it fills in the potential gaps of observing a target with a single doppler radar.

Together these two radars can observe a 3D profile of the wind vectors in the layers of the atmosphere. The calculations to create the 3D wind profile can be used as the following:

2. Radial Velocity Equations

For a given point in space, the radial velocity measured by each radar can be expressed as:

- Radar 1: $V_{r1} = u \sin \theta_1 + v \cos \theta_1 + w \sin \phi_1$
- Radar 2: $V_{r2} = u \sin \theta_2 + v \cos \theta_2 + w \sin \phi_2$

Where:

- u, v, w are the 3D wind components (east-west, north-south, and vertical).
- θ_1, θ_2 are the azimuth angles (horizontal angles) of the radars relative to the point.
- ϕ_1, ϕ_2 are the elevation angles (vertical angles) of the radars relative to the point.

Sources of retrieval uncertainty:

Signal-to-noise ratio (SNR): A dimensionless ratio of the average signal power to the power of the average noise level, used to determine the reception quality (from the backscatter from the target) and an achievable receiver sensitivity.

For radars can be calculated in a logarithmic scale: ([source, radartutorial.eu](#))

Thermodynamic Profiling Basics:

Why temperature/humidity profiles matter for stability and turbine-load context, not just wind:

Buoyancy and convection behaviors at the given time. Sharp changes or inversions of temperature and humidity profiles can be linked to unstable behaviors (and cloud convection and development) within that layer of the atmosphere, which could bring more turbulent air movement to the layer.

"High-temporal-resolution profiles of temperature and water vapor in the boundary layer are needed ... because of the potential rapid evolution of the boundary layer with time." (Turner and Lohner, 2014)

Optimal estimation theory (Rodger' framework):

The statistical backbone of TROPoe - TROPoe main workflow:

TROPoe: Tropospheric Remotely Observed Profiling via Optimal Estimation

"An optimal estimation (OE) algorithm that iteratively run a radiative transfer model and adjust its input (the thermodynamic profiles) until its output (the spectral radiance) matches the observation." (Letizia et al., 2025)

The main component of TROPoe is Optimal Estimations (OE) (Rodgers, 2000). OE is used as a method to solve the issue of ill-posed inverse problems from thermodynamic profile estimations based on observed spectral radiance at the ground level. OE allow statistic methods to estimate the most likely probability density function (p.d.f) to be compatible with both prior variable statistics (x) and current observations (y) (Stefano et al., 2025).

An example of the derived OE function is as follows (Letizia et al., 2025):

$$\underbrace{p(x|y)}_{\text{posterior}} = \underbrace{p(y|x)}_{\text{noise}} \underbrace{p(x)}_{\text{prior}} \underbrace{p(y)^{-1}}_{\text{prior normalization}}.$$

Where:

$p(x|y)$ = the forward model for atmospheric absorption and emission. Plus the instrumental noise of the of the ASSIST (Atmospheric Sander Spectrometer by Infrared Spectral Technology)

x = the thermodynamic properties of the atmosphere (at a selected height)

y = the spectral radiance and auxiliary observations

$p(x)$ = the climatology

The function above allows a complex and nonlinear system (i.e., atmospheric thermodynamics) into a more linear problem which allows a more in-depth observation into the OE theory and adding more complex components later onto the observation. (Letizia et al., 2025)

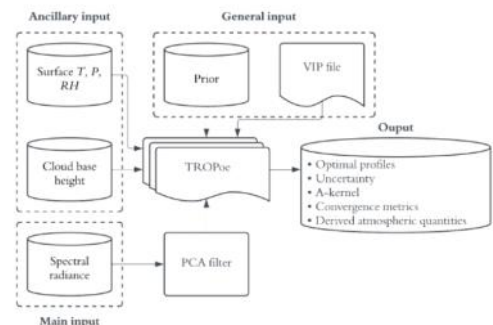


Figure 16. Flowchart of TROPoe.

$$SNR = 10 \log \frac{\text{signal power}}{\text{noise power}} = 10 \log \frac{P_S}{P_R}$$

Beam spreading:

Spreading of the transmitted optical beam in the atmosphere. Occurs when the beam is diffracted near the receiving aperture, due to atmospheric turbulence and the size of the beam is larger than the size of the eddy cells. To decrease beam spreading, you have to increase the average aperture radius (the maximum diameter of a light beam that can pass through an optical system) (Anbarasi et al., 2017).

Assumption of flow homogeneity:

A common practice for estimating the profile of the atmospheric boundary layer through doppler wind lidar (Letizia, 2024). Flow homogeneity stands means that statistically the flow within the layer of the atmosphere does not vary over time (Wilson, 2007). This simplification is used to pinpoint obstructions in the atmospheric flow of air that are either man made (wind turbines) or natural (complex terrain) (Letizia, 2024).

LiSBOA for scan optimization and LiDARGO processing:

LiSBOA: LiDAR Statistical Barnes Objective Analysis; an optimal design of lidar scans and calculations over a Cartesian grid of the statistical moments of the velocity field.

The advantage(s) of LiSBOA for lidar use usage is that it's parameterized to have the most optimal azimuthal resolutions for Lidar observations (primarily useful when using multiple lidar radars, dual-doppler or in complex terrains). The second advantage allows more control when observing turbulent or multi-variable flows from the atmosphere since LiSBOA can have a controlled response to spatial wavelengths of velocity fields in various orders. (Letizia, 2021)

LIDARGO: LiDARGO General Operation; a form of standardization and "quality control" when it comes to the analysis of raw data from scanned lidar. Fixes and has more control over the issues of position, repetition, and corrupt data (caused by either poor signal strength or saturation, or disturbances getting a read on the scanned target.) The main goal of doing this "quality control" is to allow analysis that can spot suspicious activity/data from the scanned atmospheric flow (Letizia et al., 2026).

The work flow of LIDARGO can be shown as below (Letizia et al., 2026):

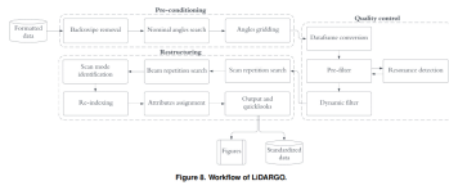


Figure 8. Workflow of LIDARGO.

For the application of wind science for turbines, LIDARGO combines static and dynamic quality flags to identify outliers in the radial wind speed (RWS) data due to a low signal-to-noise ratio (SNR) which is primarily caused by difficulties or obstructions getting a signal from the target echo.

Yes, this LIDARGO method is useful for not only standardizing a large dataset of atmospheric flow but also for bringing quality control to the data itself. However, methods like these need to be used with caution because outliers presented in the data could still be a possibility within our chaotic atmospheric system (Letizia et al., 2026).